SWE485 - Selected Topics in AI

STUDENT DEPRESSION RISK ANALYSIS

Lujain Alsudais | Hailah Alghanam | Nora Alangari | Raghad Alghamlas | Shahad Almuhaisin | Bushra Alkhulaifi | Tala Alrajeh
444200318 | 444200893 | 444202998 | 444204503 | 444200988 | 444201172 | 444200459

THE CHALLENGE & STUDENT IMPACT

THE MENTAL HEALTH CRISIS

Students face unprecedented levels of academic stress, financial pressure, and sleep deprivation. These factors often lead to undiagnosed depression, directly impacting GPA and retention rates.

NEED FOR EARLY DETECTION

Traditional counseling relies on self-reporting, which is often too late. Machine Learning offers a proactive approach by identifying risk patterns in daily habits before they escalate.

MAIN OBJECTIVES

- Build a predictive pipeline for risk analysis.
- Predict depression risk using Supervised Learning.
- Identify student archetypes via Unsupervised Learning.
- Generate AI-driven guidance for interventions.

BUSINESS VALUE

- Proactive Early Intervention.
- Data-Driven Advising.
- Scalable Personalized Support.
- Resource Optimization.

* DATA OVERVIEW

Dataset Source: Kaggle – student depression and lifestyle
Size: 100,000
Target Variable: Depression (Boolean — Binary Classification)

Variables:

- Age.
- academic pressure. (CGPA)
- sleep duration.
- study hours.
- social media hours.
- physical activity.
- stress level.

PROCESS/APPROACH

DATASET SELECTION AND PREPROCESSING

Cleaning 100k records, handling missing values, and feature scaling.

SUPERVISED LEARNING

Training Logistic Regression & Random Forest & SVM for risk prediction.

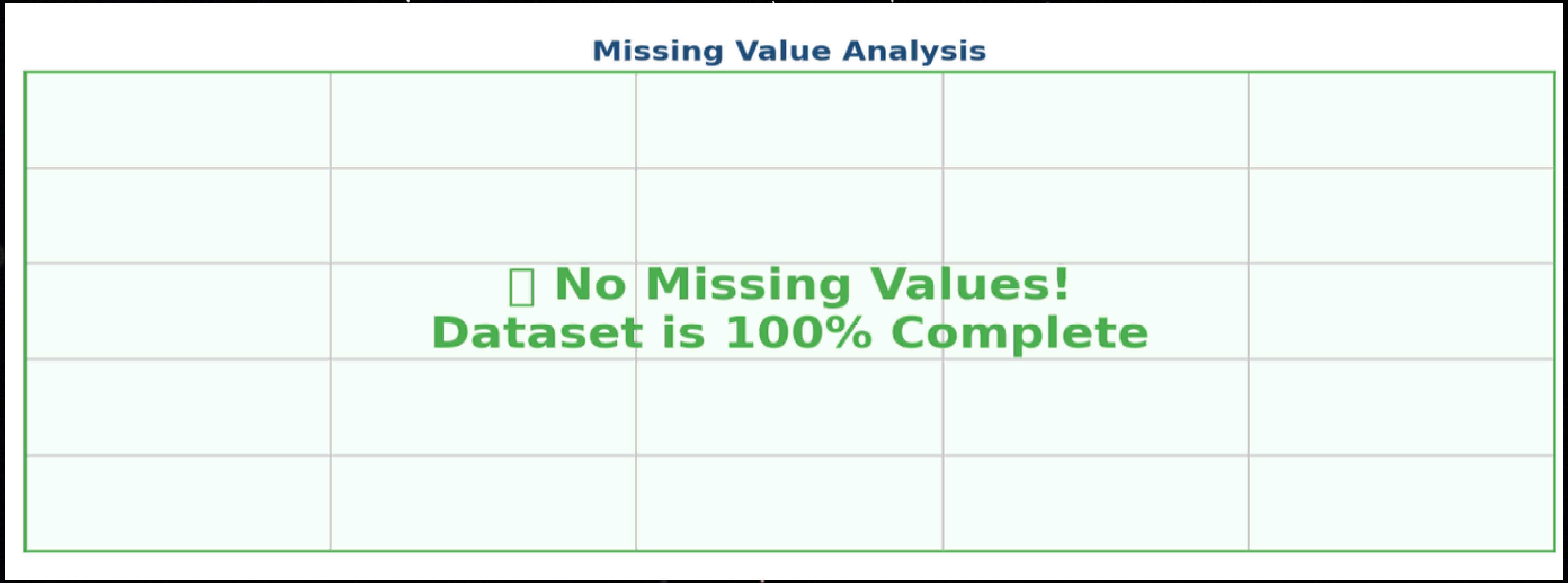
UNSUPERVISED LEARNING

KMeans and DBSCAN clustering to discover student "archetypes".

INTEGRATE GENERATIVE AI

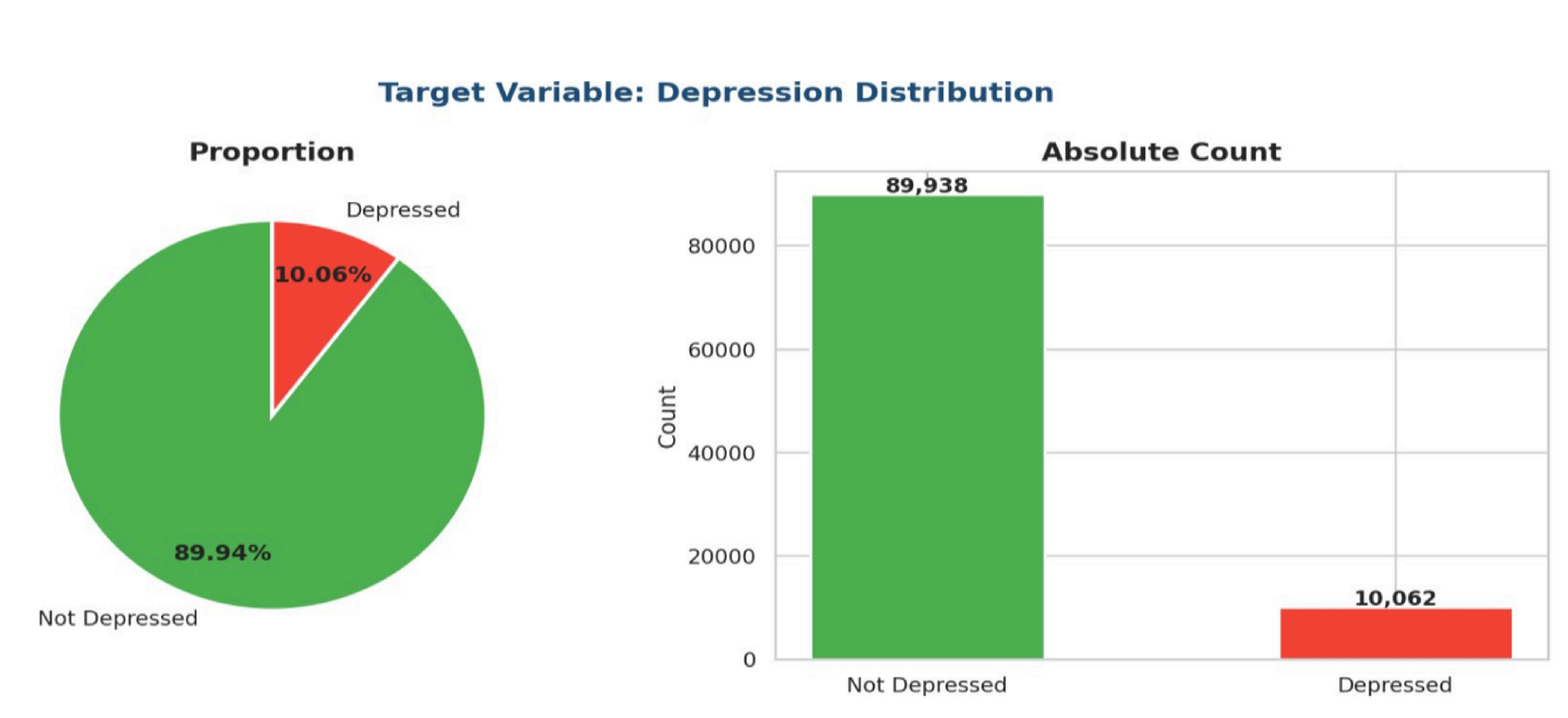
Gemini API is integrated to support the generation of natural language explanations related to depression detection.

MISSING VALUE ANALYSIS



There are no missing values in the database, and all 11 features are complete.

DEPRESSION DISTRIBUTION



Not Depressed: 89.94% (89,938)
Depressed: 10.06% (10,062)

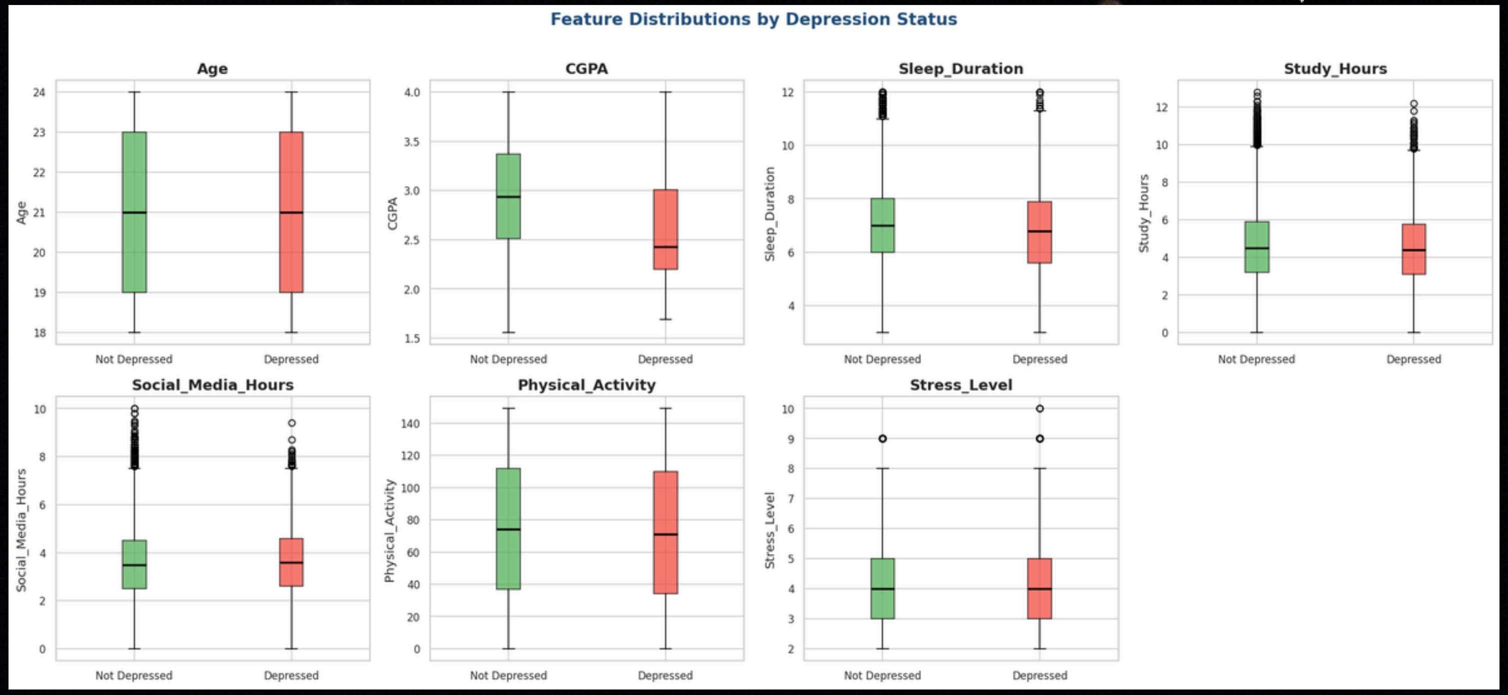
PREPROCESSING

FEATURE DISTRIBUTIONS

Key Observations :

- **CGPA:** Depressed students are clearly skewed toward lower GPA values.
- **Sleep Duration:** Non-depressed students cluster more around 7–8 hours.
- **Stress Level:** Depressed students are disproportionately represented at higher stress levels (8–10).
- **Age:** Nearly uniform across both groups.

BOX PLOTS BY DEPRESSION STATUS



Key Observations :

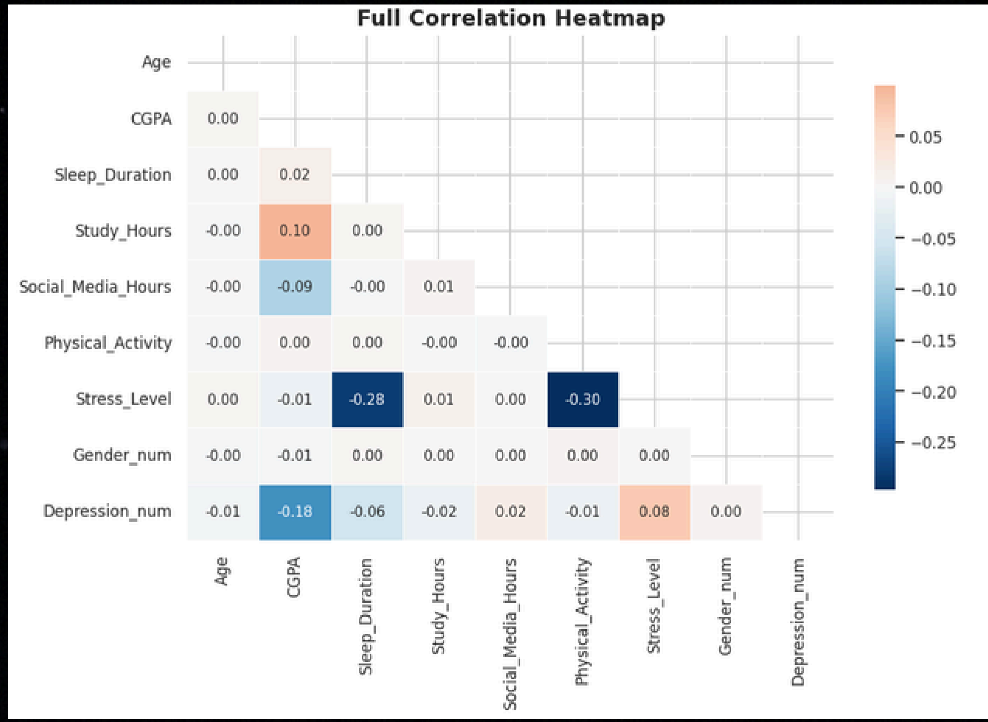
- **CGPA:** Depressed students lower (~2.5) compared to non-depressed students (~3.0).
- **Sleep Duration:** Depressed students show a lower median sleep duration and a wider spread.
- **Stress Level:** Median stress is slightly higher for depressed students.
- **Physical Activity:** Marginally lower median activity in depressed students, though overlap is large.

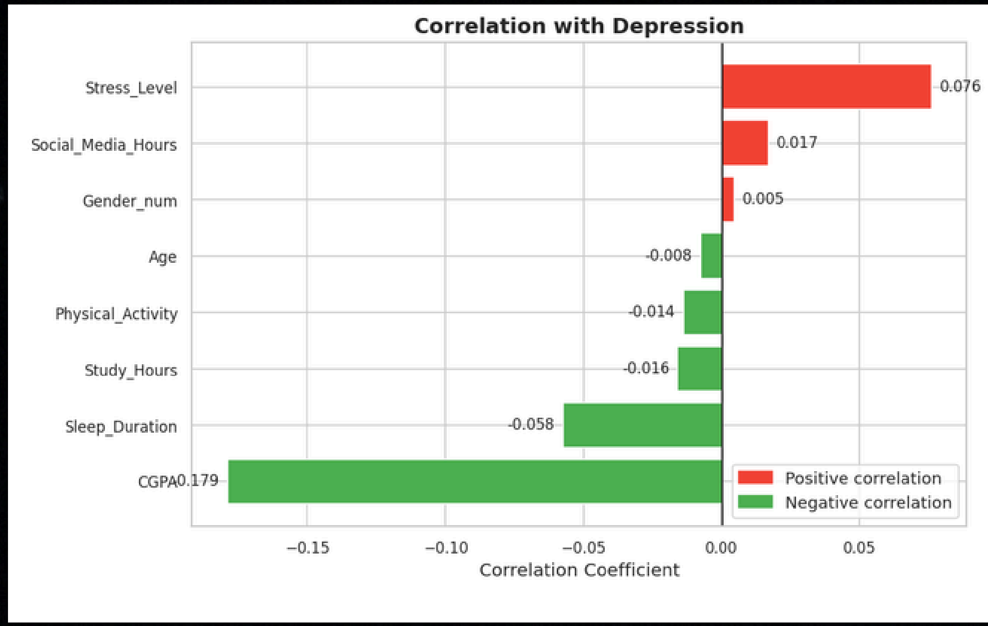
PREPROCESSING

ANALYSIS

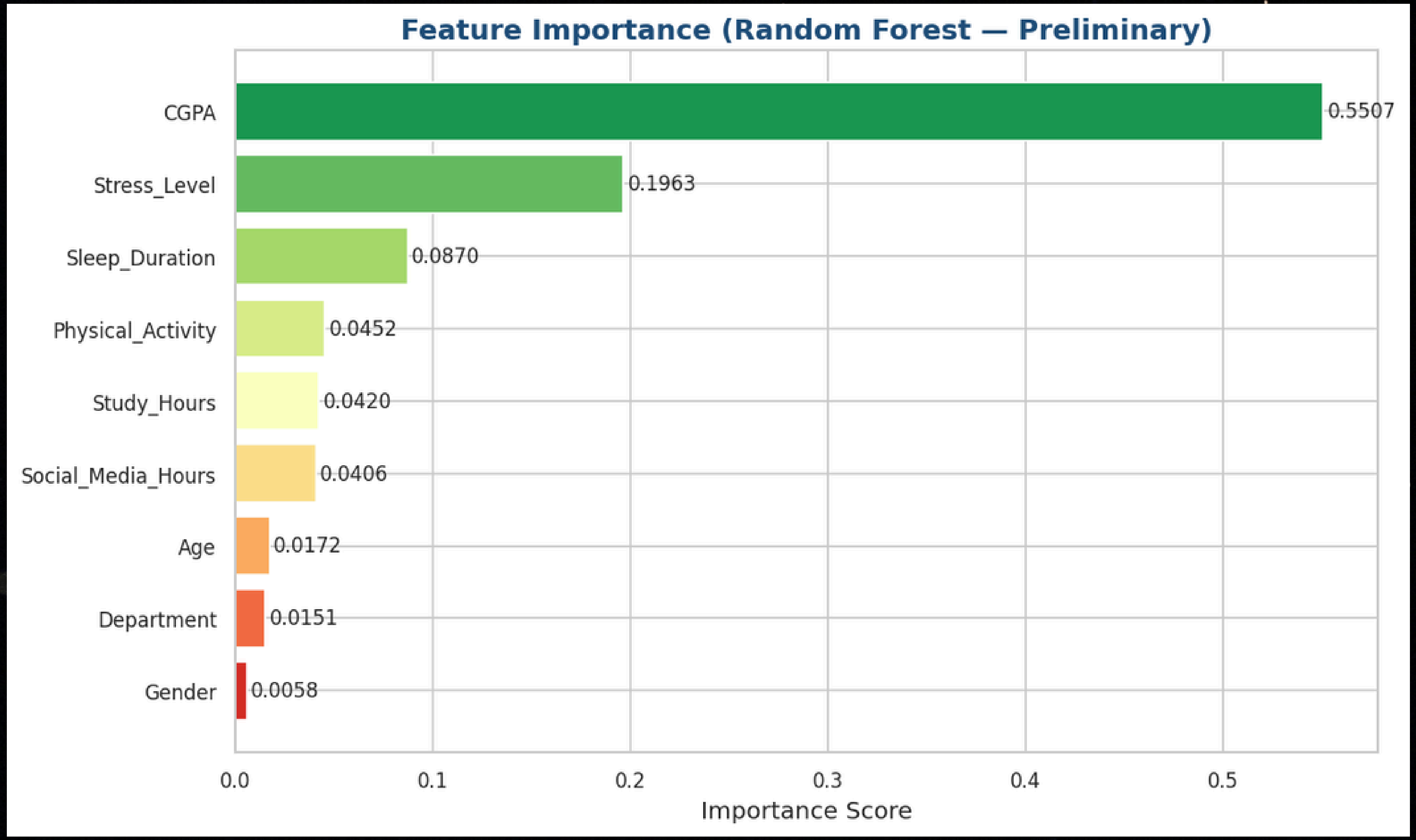
CATEGORICAL FEATURE ANALYSIS

CORRELATION ANALYSIS





FEATURE IMPORTANCE



Top 3 Most Important Features:

- CGPA: 0.5507
- Stress_Level: 0.1963
- Sleep_Duration: 0.0870

EDA SUMMARY

Total Records: 100,000
Total Features: 11
Missing Values: 0
Depressed Students : 10,062 (10.06%)
Class Imbalance : 8.9:1

Top Correlations with Depression

- CGPA r = -0.179 ↓ Negative
- Stress_Level r = +0.076 ↑ Positive
- Sleep_Duration r = -0.058 ↓ Negative
- Social_Media_Hours r = +0.017 ↑ Positive
- Study_Hours r = -0.016 ↓ Negative
- Physical_Activity r = -0.014 ↓ Negative
- Age r = -0.008 ↓ Negative
- Gender_num r = +0.005 ↑ Positive

Key Takeaways

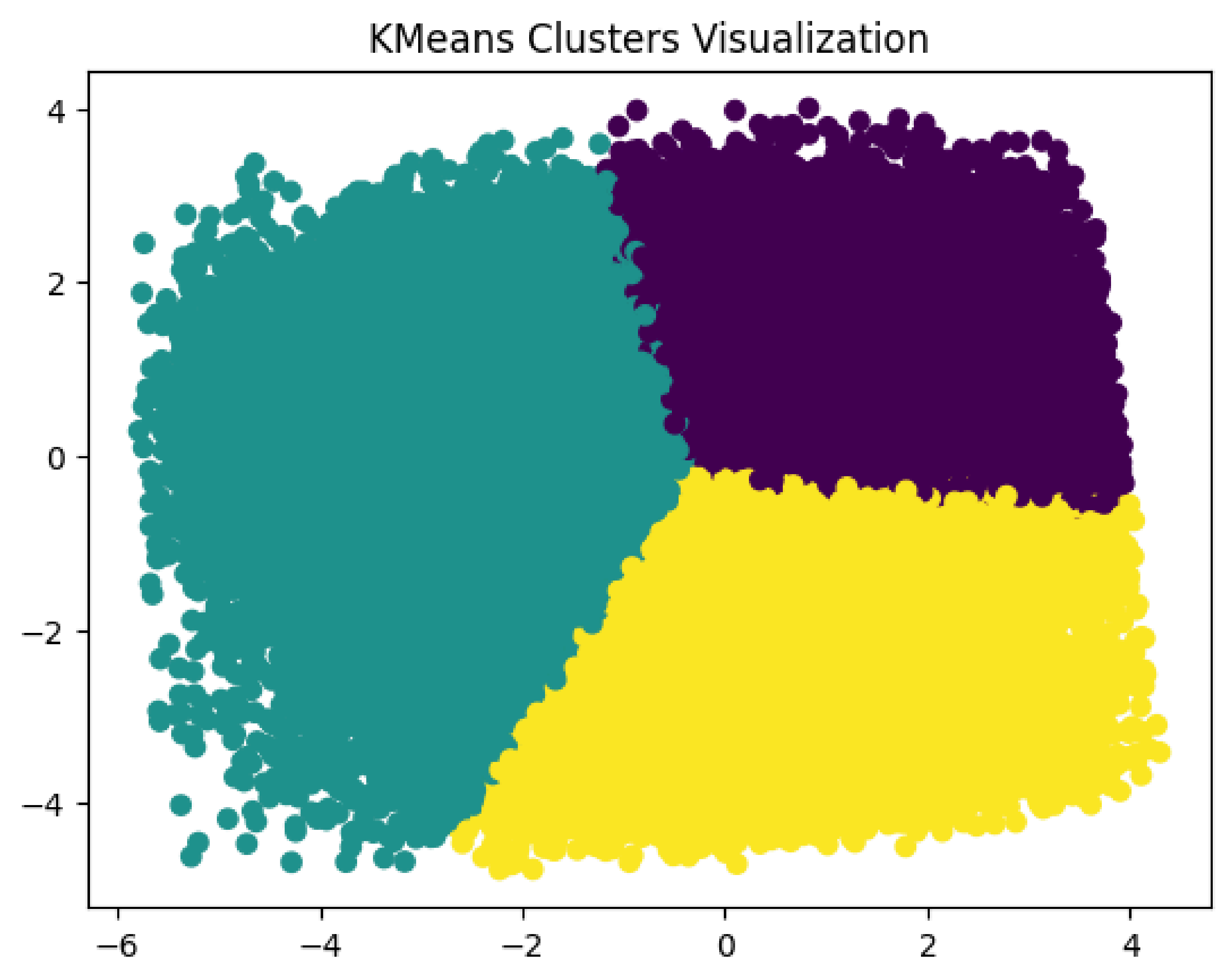
1. CGPA is the strongest predictor of depression
2. Stress_Level rises sharply to near 100% depression at level 10
3. Sleep_Duration protects against depression
4. Dataset has significant class imbalance
5. No missing values

Evaluation Metric	Random Forest (Tuned)	Logistic Regression	SVM (Linear)
Test Accuracy	**74.38%**	63.95%	63.86%
5-Fold CV Accuracy	**73.93% ± 0.30%**	63.56% ± 0.43%	63.42% ± 0.46%
ROC-AUC Score	**0.70**	N/A	N/A — LinearSVC has no `predict_proba`
Precision (Positive class)	**0.21**	0.16	0.16
Recall (Positive class)	**0.62**	0.69	0.68
F1-Score (Positive class)	**0.33**	0.27	0.27

Random Forest demonstrated superior capability in capturing complex behavioral patterns in the student lifestyle data compared to linear models.

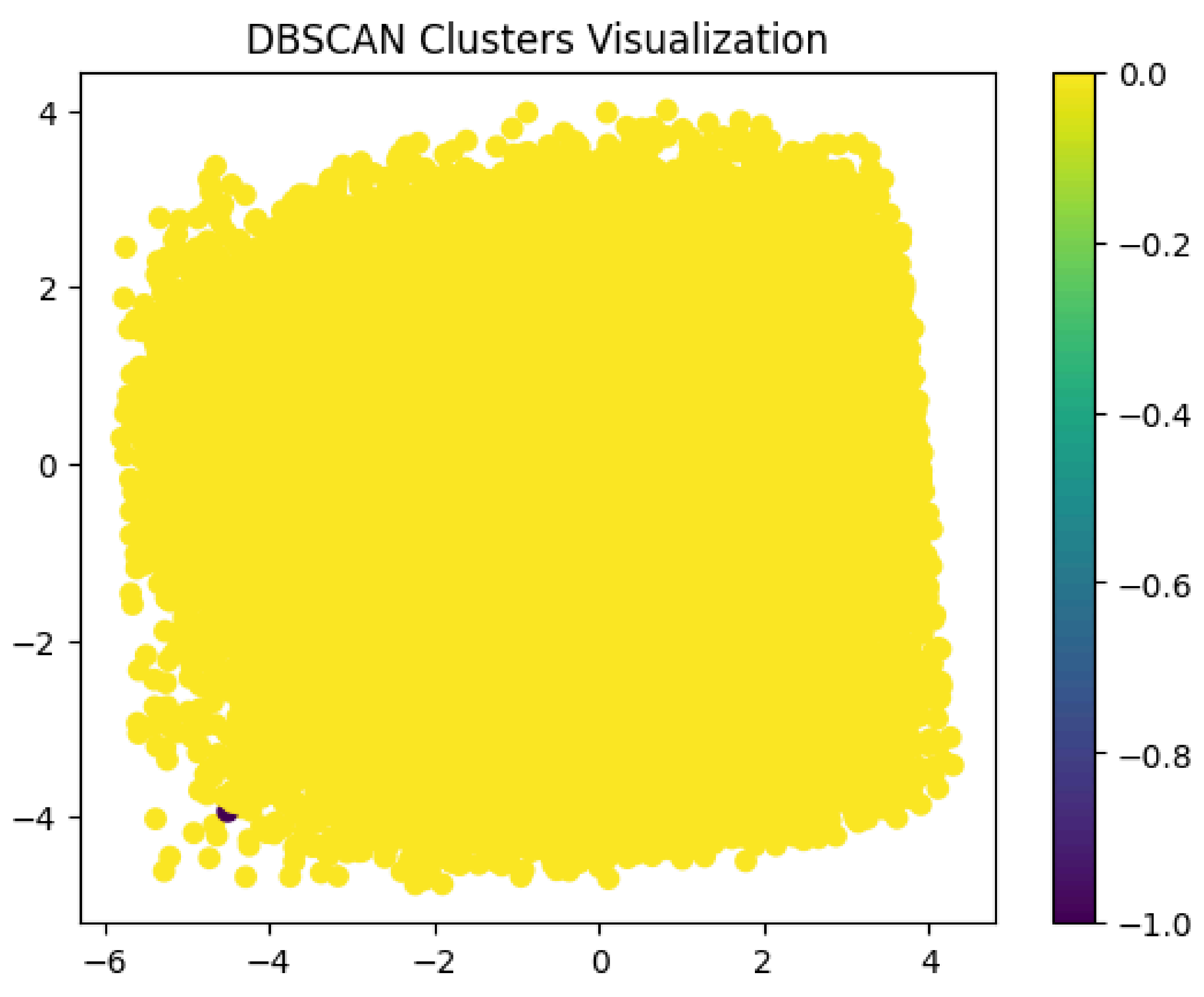
UNSUPERVISED LEARNING

K Means



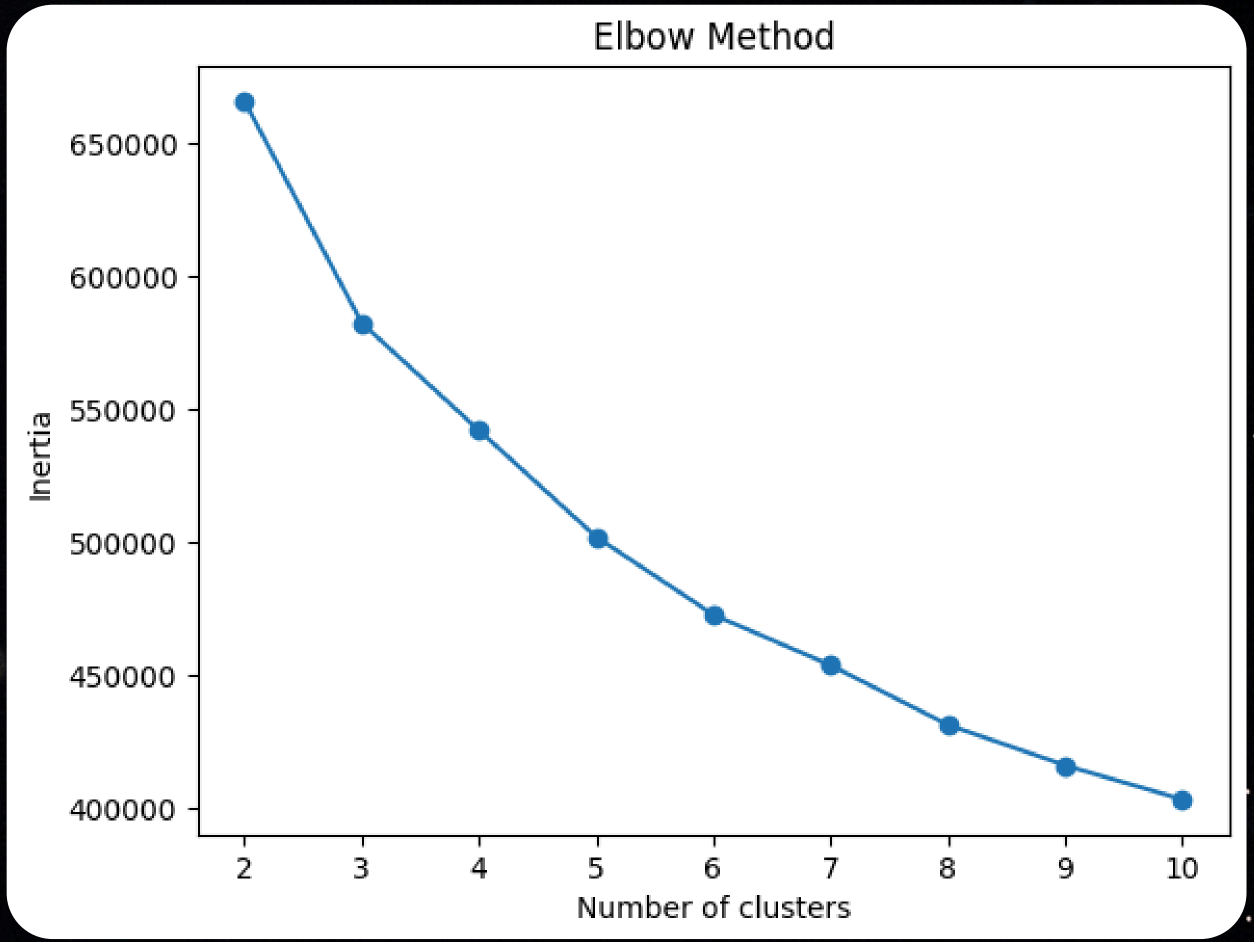
Cluster	Label	Depression Rate	Risk Level
0	Sedentary Low-Study	9.67%	Moderate
1	Balanced Active	8.99%	Low (reference)
2	Sleep-Deprived High-Stress	10.92%	High

DBSCAN

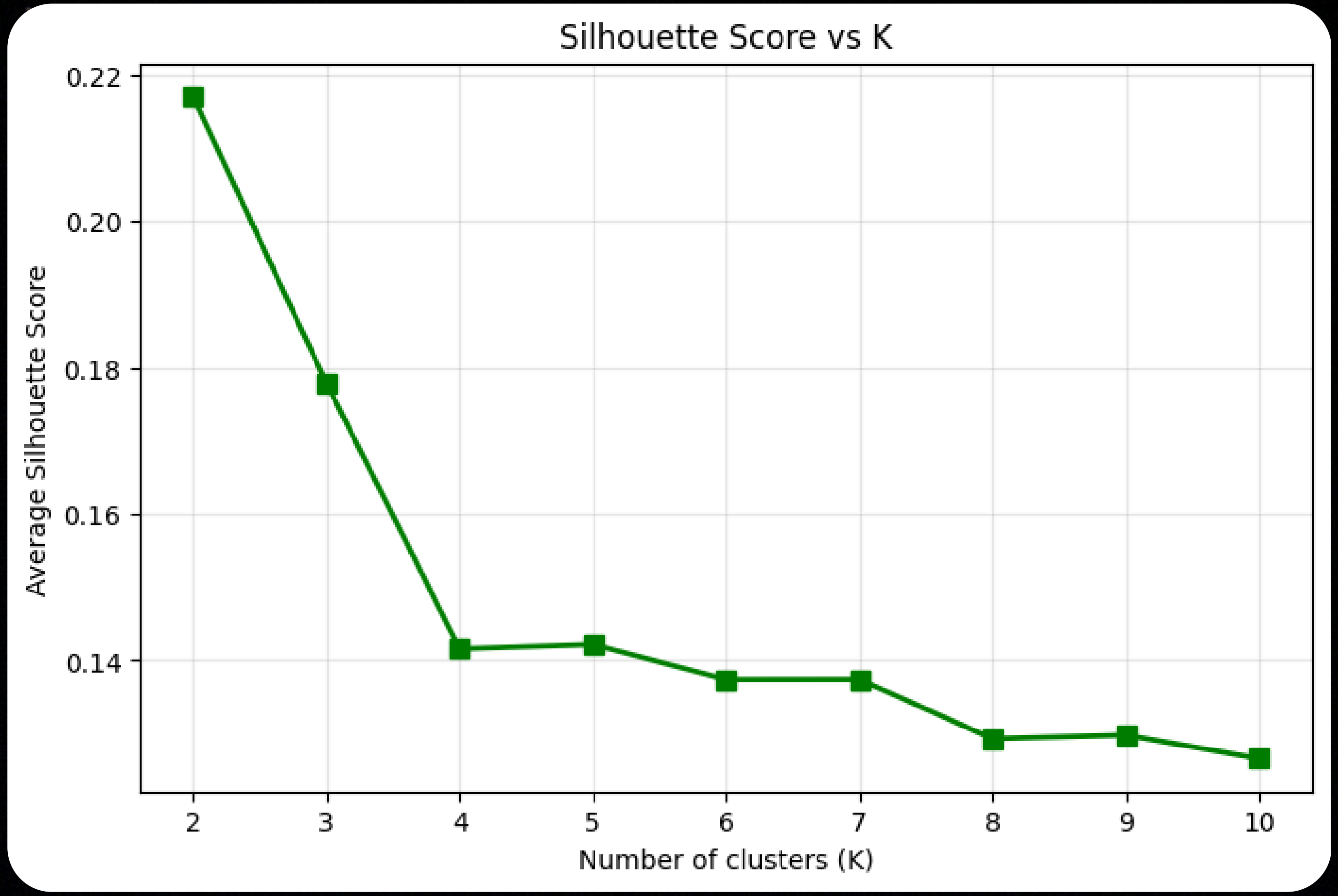


eps=1.5 collapsed almost all 55,004 students into a single cluster (only 2 noise points).
there are no truly isolated high-density islands so it is not suitable for integration in this dataset.

UNSUPERVISED LEARNING



“how compact clusters are”



“how separated clusters are ”

TEMPLATE 1:

You are an assistant that provides simple student lifestyle advice.
Student information:
CGPA: 2.1 Sleep Duration: 4 hours Study Hours: 10 hours Social Media Hours: 5 hours Physical Activity: 0 hours Stress Level: 9 Prediction Result: At risk of depression
Based on the prediction result, explain the student's situation and give simple advice.
Do not claim that the student is clinically depressed. Treat the prediction as a risk indicator only.
Keep the response clear and easy to understand.

TEMPLATE 2:

You are a mental health and academic advisor.
Student information:
CGPA: 2.1 Sleep Duration: 4 hours Study Hours: 10 hours Social Media Hours: 5 hours Physical Activity: 0 hours Stress Level: 9 Prediction Result: At risk of depression
Provide: Explanation - Risk factors - Lifestyle recommendations - Study balance advice
Safety instruction: Do not claim that the student is clinically depressed. Treat the prediction as a risk indicator only.

TEMPLATE 3:

You are a supportive student coach.
Student information:
CGPA: 2.1 Sleep Duration: 4 hours Study Hours: 10 hours Social Media Hours: 5 hours Physical Activity: 0 hours Stress Level: 9 Prediction Result: At risk of depression
Give: Personalized advice - Daily action steps - Encouragement
Safety instruction: Do not claim that the student is clinically depressed. Treat the prediction as a risk indicator only.

TEMPLATE 4:

You are a data-driven advisor.
Student information:
CGPA: 2.1 Sleep Duration: 4 hours Study Hours: 10 hours Social Media Hours: 5 hours Physical Activity: 0 hours Stress Level: 9 Prediction Result: At risk of depression
Explain: Why this prediction happened - Which factors affected it - What should be improved first
Safety instruction: Do not claim that the student is clinically depressed. Treat the prediction as a risk indicator only.

TEMPLATE PERFORMANCE SUMMARY

Template	Clarity	Relevance	Personalization	Completeness	Safety	Best Suited For
T1: Basic	3 / 4	3 / 4	2 / 4	2 / 4	4 / 4	Quick, simple advice for general audiences
T2: Structured	4 / 4	4 / 4	3 / 4	4 / 4	4 / 4	Comprehensive reports with consistent formatting
T3: Personalized	3 / 4	4 / 4	4 / 4	3 / 4	4 / 4	Student engagement, motivation, and coaching
T4: Analytical	4 / 4	4 / 4	3 / 4	3 / 4	4 / 4	Data-driven explanations and interpretability

CONCLUSION

Early Detection

ML models like Random Forest can identify risks with ~75% accuracy before symptoms become clinical.

Sleep is Key

Cluster 2 shows that sleep deprivation is the most significant lifestyle factor correlated with higher depression risk.

AI Intervention

Generative AI provides the necessary bridge between predictive analytics and human-centric intervention strategies.

THANK YOU.